

Education

PhD in Mathematics <i>Advisor:</i> Michael O'Neil <i>Thesis:</i> Fast transform methods for Gaussian random fields	<i>Courant Institute, New York University</i>	2020-2025
BS in Computational and Applied Mathematics <i>Advisor:</i> Mihai Anitescu <i>Thesis:</i> Nonstationary Gaussian process approximations of piecewise analytic computer codes	<i>University of Chicago</i>	2015-2019

Research Positions

Peter O'Donnell Jr. Postdoctoral Fellow <i>Advisors:</i> Joe Kileel, Per-Gunnar Martinsson	<i>Oden Institute, UT Austin</i>	2025-present
CSGF Practicum <i>Advisors:</i> Yang Liu, Xiaoye Sherry Li	<i>Lawrence Berkeley National Laboratory</i>	2023
Mathematics and Statistics Researcher <i>Advisors:</i> Mihai Anitescu, Michael Stein	<i>Argonne National Laboratory</i>	2019-2020

Publications

Persson, David, **Paul G. Beckman**, Tyler Chen, Diana Halikias, Christopher Musco. "A Recursive Butterfly Factorization with Optimality Guarantees." arXiv preprint arXiv:2607.29361 (2026).

Beckman, Paul G., Samuel F. Potter, Michael O'Neil. "A Butterfly-Accelerated Manifold Harmonic Transform." arXiv preprint arXiv:2605.21828 (2026).

Beckman, Paul G., Michael O'Neil. "A Nonuniform Fast Hankel Transform." *SIAM Journal on Scientific Computing*, no. 2 (2026): A784-A803.

Geoga, Christopher J., **Beckman, Paul G.**. "Nonparametric spectral density estimation from irregularly sampled data." arXiv preprint arXiv: 2503.00492 (2025).

Beckman, Paul G., Christopher J. Geoga. "Fast Adaptive Fourier Integration for Spectral Densities of Gaussian Processes." *Statistics and Computing* 34, no. 6 (2024): 217.

Beckman, Paul G., Christopher J. Geoga, Michael L. Stein, and Mihai Anitescu. "Scalable Computations for Nonstationary Gaussian Processes." *Statistics and Computing* 33, no. 4 (2023): 84.

Williams-Young, David B., **Paul G. Beckman**, and Chao Yang. "A Shift Selection Strategy for Parallel Shift-Invert Spectrum Slicing in Symmetric Self-Consistent Eigenvalue Computation." *ACM Transactions on Mathematical Software (TOMS)* 46, no. 4 (2020): 1-31.

Awards

Moses A. Greenfield Research Prize	<i>Courant Institute</i>	2024
Computational Science Graduate Fellowship	<i>Department of Energy</i>	2020

Talks

BIRS "Fast transforms for Gaussian random fields"	<i>Workshop on Kernel Approximation and Gaussian Processes</i>	2026
CodEx Seminar "Fast transforms for Gaussian random fields"	<i>CodEx Seminar</i>	2026
UT Austin "Fast transforms for Gaussian random fields"	<i>Oden Institute Seminar</i>	2026

UT Austin	<i>Data & Algebra Seminar</i>	2026
"Implicit tensor moment estimation of nonparametric mixture models with sparse dependence"		
SIAM	<i>Uncertainty Quantification</i>	2026
"Fast transforms for spectral modeling of Gaussian processes"		
SIAM	<i>Computational Science and Engineering (Minisymposium co-organizer)</i>	2025
"Fast adaptive Fourier integration of spectral densities"		
Princeton University	<i>PACM IDeAS Seminar</i>	2025
"A nonuniform fast Hankel transform"		
SIAM	<i>Uncertainty Quantification</i>	2024
Talk: "Fast adaptive Fourier integration of spectral densities"		
Poster: "Butterfly-accelerated Gaussian random fields on manifolds"		
ICIAM	<i>International Congress on Industrial and Applied Mathematics</i>	2023
"Boundary integral methods for computing covariances in inverse source problems"		
SIAM	<i>Mathematics of Data Science (Minisymposium co-organizer)</i>	2022
"Fast algorithms for elliptic PDEs with Gaussian boundary noise"		

Teaching

Diff. Eq. with Linear Algebra	<i>UT Austin M427J</i>	Instructor	Fall 2026
Preparation for Calculus	<i>UT Austin M305G (TPEI)</i>	Instructor	Fall 2026
Differential Calculus	<i>UT Austin M408K (TPEI)</i>	Co-instructor	Spring 2026
Mathematic Statistics	<i>NYU MATH-UA.2340</i>	Teaching Assistant	Spring 2024
Statistics	<i>NYU MATH-GA.2962</i>	Grader	Fall 2021
Computational Statistics	<i>NYU MATH-GA.2080</i>	Grader	Spring 2021

Software

Primary developer	FastHankelTransform.jl, SpectralKernels.jl
Contributor	FastGaussQuadrature.jl, chunkIE

Outreach and Service

Peer reviewer

IEEE Signal Processing Letters, Statistics and Computing, Journal of Machine Learning Research, SIAM Journal on Scientific Computing.

Volunteer Instructor	<i>Texas Prison Education Initiative (TPEI)</i>	2026-present
Instructor for credit-bearing college math courses for incarcerated youth and adults		
Volunteer Instructor	<i>Petey Greene Program</i>	2020-2025
Instructor for elementary through college level math for currently and formerly incarcerated adults		
Reading Group Founder	<i>Courant Institute</i>	2020-2024
Co-founded a department group reading articles on progress towards more accessible math higher education		
Department Climate Survey Designer	<i>Courant Institute</i>	2020-2022
Co-designed, administered, and reviewed a survey on Courant Institute PhD student experience		